

# Sequential Testing: ROPE vs Bayes Factor vs LOO (Continuous)

Understanding Divergences in Bayesian Decision Making

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## 1 Part 2: Continuous Predictor Simulation

In this second part, we repeat the sequential testing process for a **continuous predictor** (e.g., word frequency or trial number) to see if decision metrics behave similarly.

### 1.1 Data Generation (Continuous)

We simulate a log-RT variable with a continuous predictor  $X$  (centered, range approx -2 to 2). True Effect of  $X$ : 0.05 (approx 5% change per unit of  $X$ ).

### 1.2 Simulation Loop (Continuous)

We fit three models again: 1. **Null**:  $y \sim 1 + (1|\text{subject}) + (1|\text{item})$  2. **Wide**:  $y \sim \text{trial\_x}$ , prior  $\text{normal}(0, 5)$  3. **Narrow**:  $y \sim \text{trial\_x}$ , prior  $\text{normal}(0, 0.2)$

### 1.3 Continuous Results Table

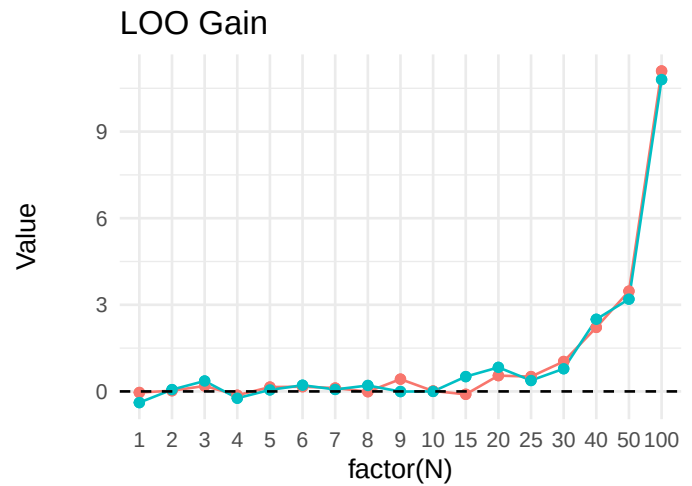
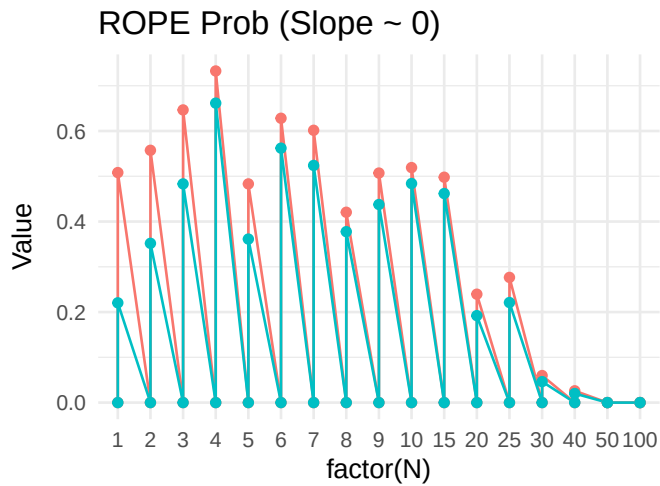
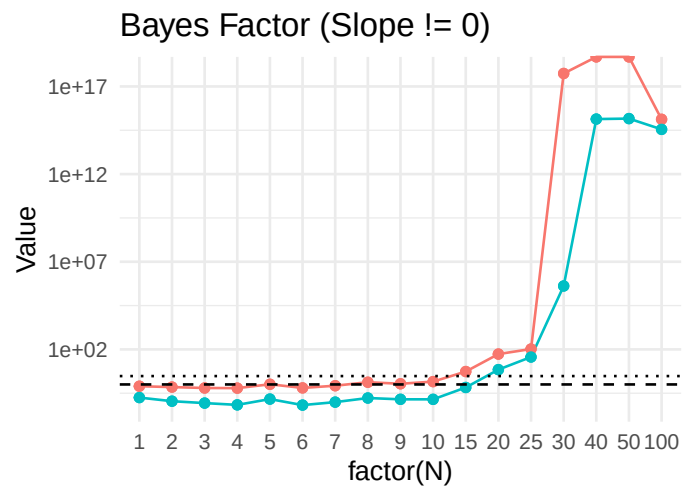
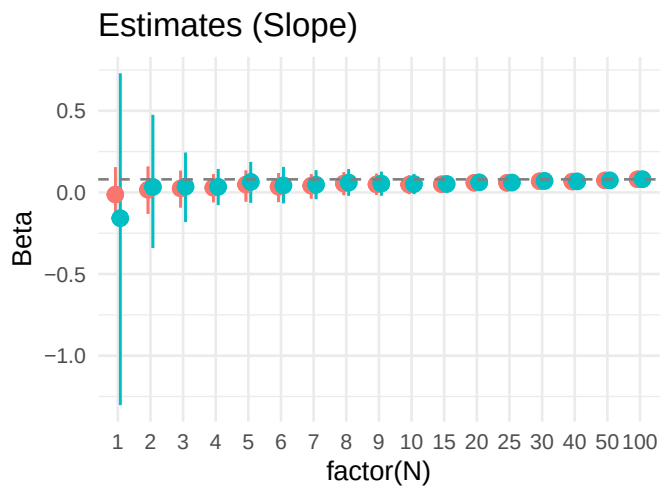
Table 1: Continuous Predictor: Decision Metrics

| N | Prior  | Estimate_CI         | BF10         | ROPE_Prob | LOO_Gain |
|---|--------|---------------------|--------------|-----------|----------|
| 1 | Wide   | -0.16 [-1.30, 0.73] | 1.780000e-01 | 0.000     | -0.387   |
| 1 | Wide   | -0.16 [-1.30, 0.73] | 1.780000e-01 | 0.220     | -0.387   |
| 1 | Narrow | -0.01 [-0.16, 0.15] | 7.920000e-01 | 0.000     | -0.038   |
| 1 | Narrow | -0.01 [-0.16, 0.15] | 7.920000e-01 | 0.508     | -0.038   |
| 2 | Wide   | 0.03 [-0.34, 0.47]  | 1.110000e-01 | 0.000     | 0.062    |
| 2 | Wide   | 0.03 [-0.34, 0.47]  | 1.110000e-01 | 0.352     | 0.062    |
| 2 | Narrow | 0.02 [-0.13, 0.16]  | 7.070000e-01 | 0.000     | 0.023    |

| N  | Prior  | Estimate_CI        | BF10         | ROPE_Prob | LOO_Gain |
|----|--------|--------------------|--------------|-----------|----------|
| 2  | Narrow | 0.02 [-0.13, 0.16] | 7.070000e-01 | 0.557     | 0.023    |
| 3  | Wide   | 0.03 [-0.18, 0.24] | 8.600000e-02 | 0.000     | 0.361    |
| 3  | Wide   | 0.03 [-0.18, 0.24] | 8.600000e-02 | 0.483     | 0.361    |
| 3  | Narrow | 0.02 [-0.09, 0.13] | 6.280000e-01 | 0.000     | 0.190    |
| 3  | Narrow | 0.02 [-0.09, 0.13] | 6.280000e-01 | 0.647     | 0.190    |
| 4  | Wide   | 0.03 [-0.08, 0.14] | 6.800000e-02 | 0.000     | -0.235   |
| 4  | Wide   | 0.03 [-0.08, 0.14] | 6.800000e-02 | 0.662     | -0.235   |
| 4  | Narrow | 0.03 [-0.06, 0.11] | 6.170000e-01 | 0.000     | -0.133   |
| 4  | Narrow | 0.03 [-0.06, 0.11] | 6.170000e-01 | 0.733     | -0.133   |
| 5  | Wide   | 0.06 [-0.06, 0.19] | 1.460000e-01 | 0.000     | 0.053    |
| 5  | Wide   | 0.06 [-0.06, 0.19] | 1.460000e-01 | 0.361     | 0.053    |
| 5  | Narrow | 0.05 [-0.06, 0.13] | 1.023000e+00 | 0.000     | 0.150    |
| 5  | Narrow | 0.05 [-0.06, 0.13] | 1.023000e+00 | 0.483     | 0.150    |
| 6  | Wide   | 0.04 [-0.07, 0.16] | 6.600000e-02 | 0.000     | 0.214    |
| 6  | Wide   | 0.04 [-0.07, 0.16] | 6.600000e-02 | 0.562     | 0.214    |
| 6  | Narrow | 0.04 [-0.06, 0.12] | 6.350000e-01 | 0.000     | 0.163    |
| 6  | Narrow | 0.04 [-0.06, 0.12] | 6.350000e-01 | 0.628     | 0.163    |
| 7  | Wide   | 0.05 [-0.04, 0.14] | 9.800000e-02 | 0.000     | 0.069    |
| 7  | Wide   | 0.05 [-0.04, 0.14] | 9.800000e-02 | 0.524     | 0.069    |
| 7  | Narrow | 0.04 [-0.04, 0.11] | 8.460000e-01 | 0.000     | 0.117    |
| 7  | Narrow | 0.04 [-0.04, 0.11] | 8.460000e-01 | 0.602     | 0.117    |
| 8  | Wide   | 0.06 [-0.02, 0.14] | 1.680000e-01 | 0.000     | 0.205    |
| 8  | Wide   | 0.06 [-0.02, 0.14] | 1.680000e-01 | 0.378     | 0.205    |
| 8  | Narrow | 0.05 [-0.02, 0.12] | 1.338000e+00 | 0.000     | -0.010   |
| 8  | Narrow | 0.05 [-0.02, 0.12] | 1.338000e+00 | 0.421     | -0.010   |
| 9  | Wide   | 0.05 [-0.02, 0.13] | 1.420000e-01 | 0.000     | -0.005   |
| 9  | Wide   | 0.05 [-0.02, 0.13] | 1.420000e-01 | 0.438     | -0.005   |
| 9  | Narrow | 0.05 [-0.02, 0.11] | 1.094000e+00 | 0.000     | 0.426    |
| 9  | Narrow | 0.05 [-0.02, 0.11] | 1.094000e+00 | 0.507     | 0.426    |
| 10 | Wide   | 0.05 [-0.01, 0.11] | 1.410000e-01 | 0.000     | 0.001    |
| 10 | Wide   | 0.05 [-0.01, 0.11] | 1.410000e-01 | 0.484     | 0.001    |
| 10 | Narrow | 0.05 [-0.01, 0.10] | 1.450000e+00 | 0.000     | 0.014    |
| 10 | Narrow | 0.05 [-0.01, 0.10] | 1.450000e+00 | 0.519     | 0.014    |
| 15 | Wide   | 0.05 [0.01, 0.09]  | 6.710000e-01 | 0.000     | 0.513    |
| 15 | Wide   | 0.05 [0.01, 0.09]  | 6.710000e-01 | 0.462     | 0.513    |
| 15 | Narrow | 0.05 [0.01, 0.09]  | 5.445000e+00 | 0.000     | -0.100   |
| 15 | Narrow | 0.05 [0.01, 0.09]  | 5.445000e+00 | 0.498     | -0.100   |
| 20 | Wide   | 0.06 [0.03, 0.09]  | 7.014000e+00 | 0.000     | 0.833    |
| 20 | Wide   | 0.06 [0.03, 0.09]  | 7.014000e+00 | 0.192     | 0.833    |
| 20 | Narrow | 0.06 [0.03, 0.09]  | 5.400300e+01 | 0.000     | 0.548    |
| 20 | Narrow | 0.06 [0.03, 0.09]  | 5.400300e+01 | 0.239     | 0.548    |
| 25 | Wide   | 0.06 [0.03, 0.09]  | 3.674800e+01 | 0.000     | 0.379    |
| 25 | Wide   | 0.06 [0.03, 0.09]  | 3.674800e+01 | 0.221     | 0.379    |
| 25 | Narrow | 0.06 [0.03, 0.09]  | 1.050360e+02 | 0.000     | 0.513    |
| 25 | Narrow | 0.06 [0.03, 0.09]  | 1.050360e+02 | 0.277     | 0.513    |
| 30 | Wide   | 0.07 [0.04, 0.10]  | 4.132826e+05 | 0.000     | 0.786    |

| N   | Prior  | Estimate_CI       | BF10         | ROPE_Prob | LOO_Gain |
|-----|--------|-------------------|--------------|-----------|----------|
| 30  | Wide   | 0.07 [0.04, 0.10] | 4.132826e+05 | 0.046     | 0.786    |
| 30  | Narrow | 0.07 [0.04, 0.09] | 5.481336e+17 | 0.000     | 1.036    |
| 30  | Narrow | 0.07 [0.04, 0.09] | 5.481336e+17 | 0.060     | 1.036    |
| 40  | Wide   | 0.07 [0.05, 0.09] | 1.375817e+15 | 0.000     | 2.498    |
| 40  | Wide   | 0.07 [0.05, 0.09] | 1.375817e+15 | 0.020     | 2.498    |
| 40  | Narrow | 0.07 [0.05, 0.09] | Inf          | 0.000     | 2.218    |
| 40  | Narrow | 0.07 [0.05, 0.09] | Inf          | 0.026     | 2.218    |
| 50  | Wide   | 0.07 [0.06, 0.09] | 1.465803e+15 | 0.000     | 3.197    |
| 50  | Wide   | 0.07 [0.06, 0.09] | 1.465803e+15 | 0.000     | 3.197    |
| 50  | Narrow | 0.07 [0.06, 0.09] | Inf          | 0.000     | 3.473    |
| 50  | Narrow | 0.07 [0.06, 0.09] | Inf          | 0.000     | 3.473    |
| 100 | Wide   | 0.08 [0.07, 0.09] | 3.552615e+14 | 0.000     | 10.803   |
| 100 | Wide   | 0.08 [0.07, 0.09] | 3.552615e+14 | 0.000     | 10.803   |
| 100 | Narrow | 0.08 [0.07, 0.09] | 1.330036e+15 | 0.000     | 11.100   |
| 100 | Narrow | 0.08 [0.07, 0.09] | 1.330036e+15 | 0.000     | 11.100   |

## 1.4 Visualization (Continuous)



Prior ● Narrow ● Wide

Prior — Narrow — Wide